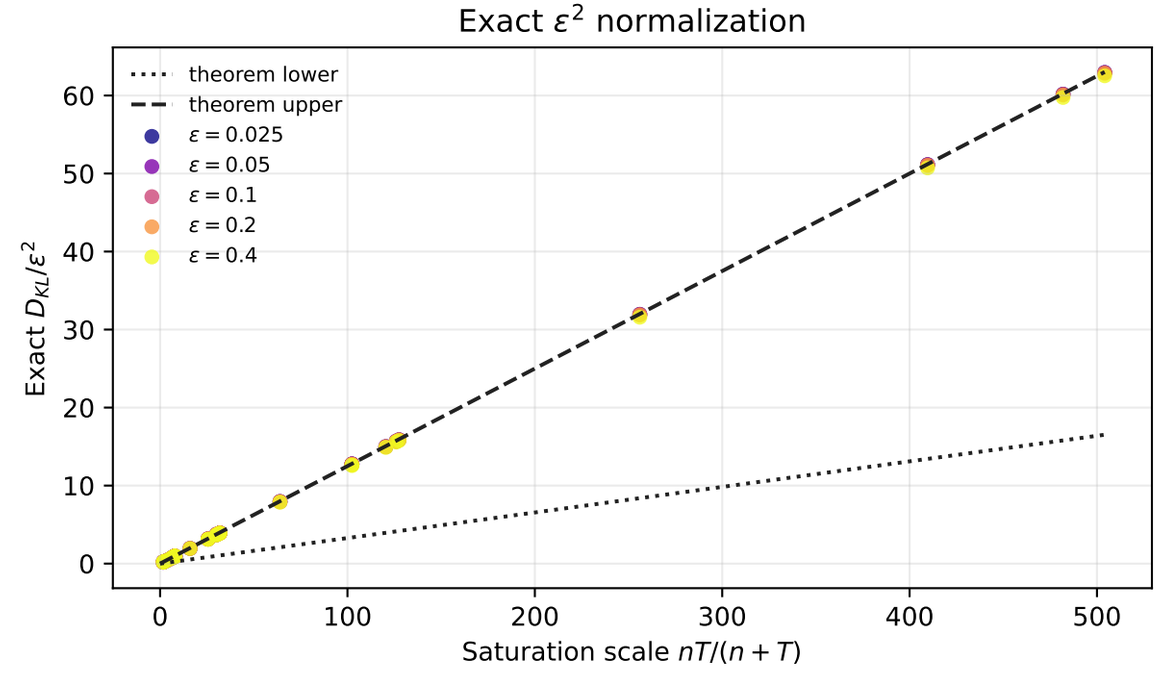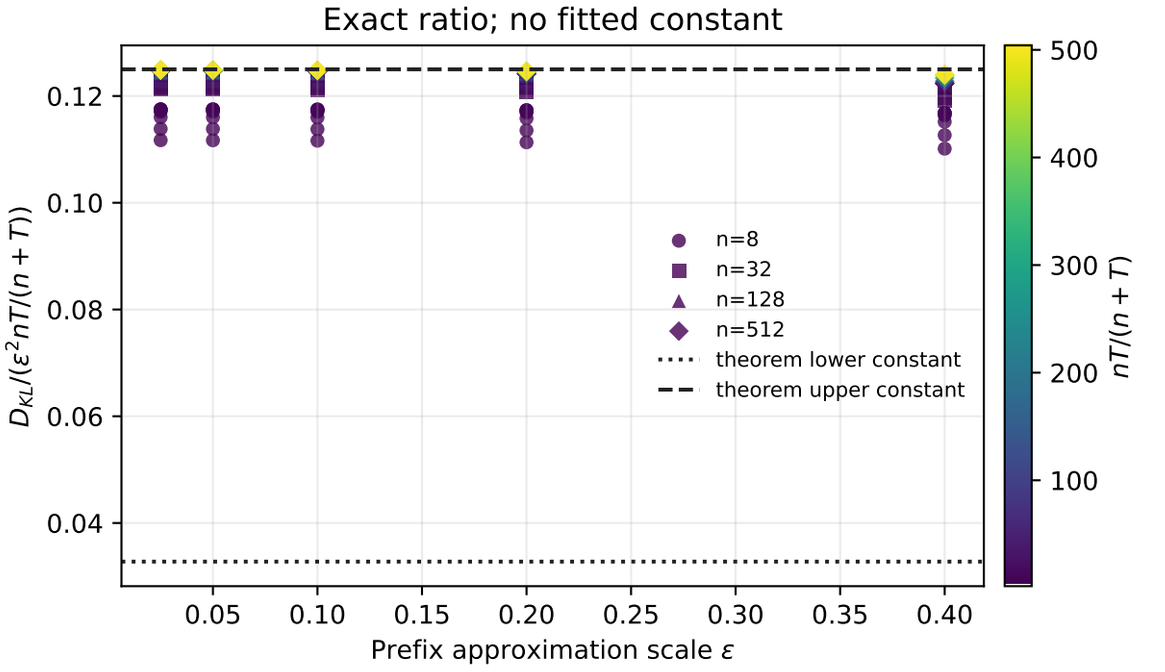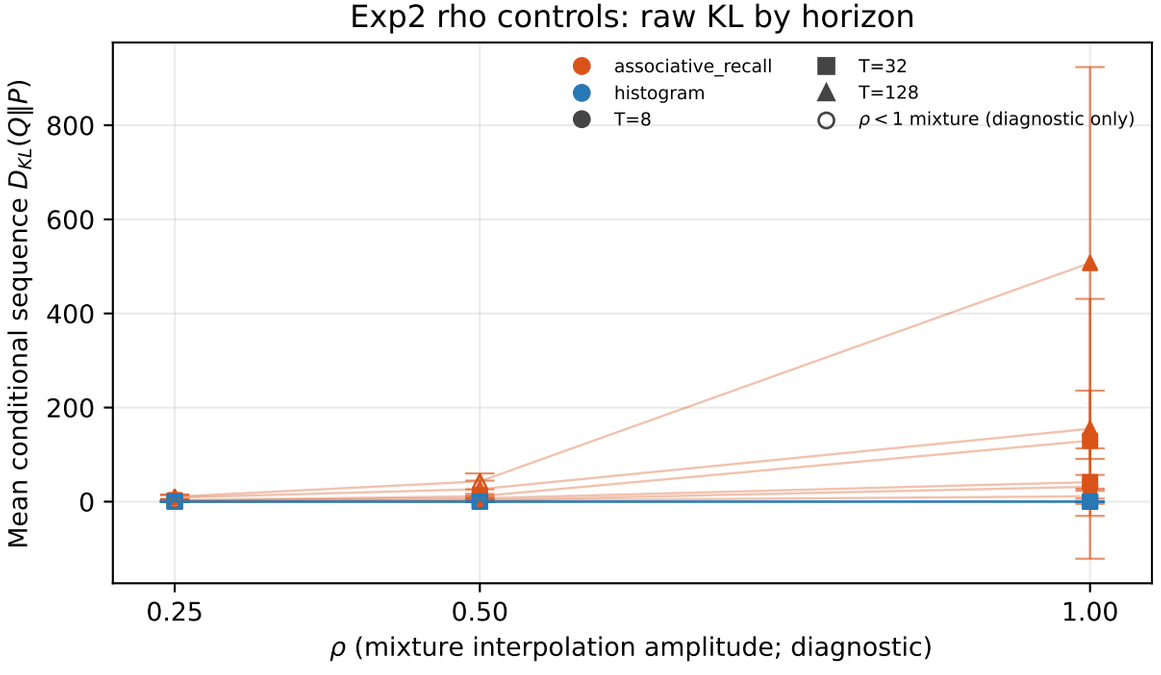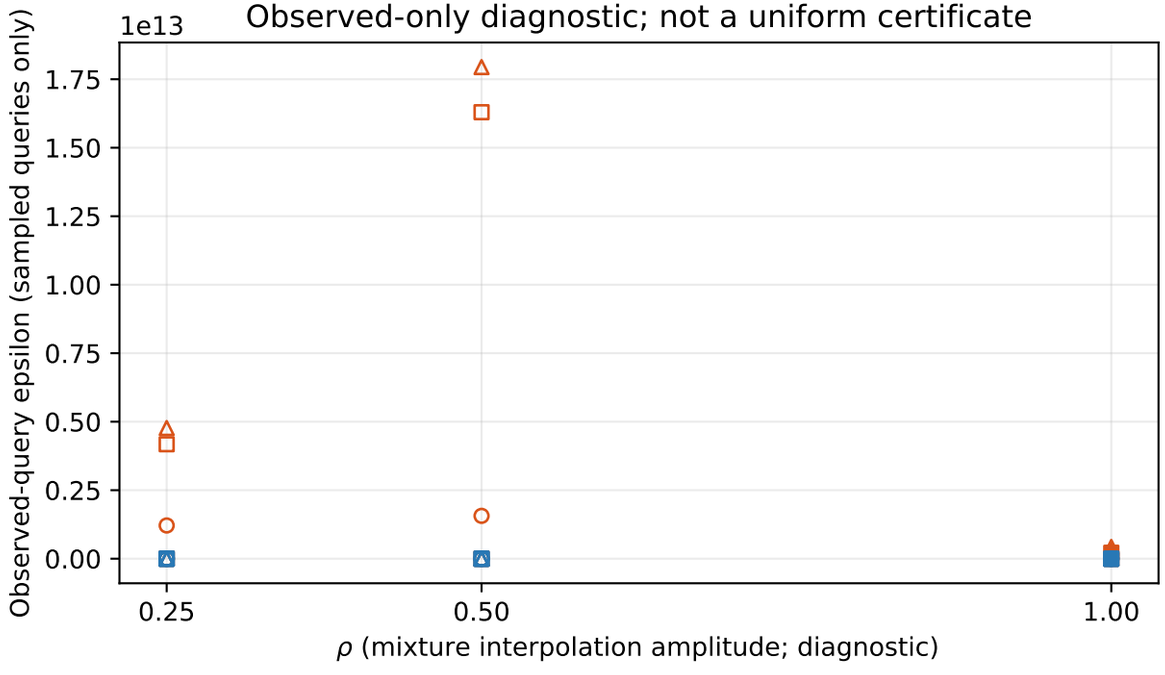Experiment 1 exact precision scaling and Exp2 rho diagnostics